Question ID: 16de54c7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question

$2x^2 + 5x - 12$

If the given expression is rewritten in the form $(2x-3)(x+k)$, where k is a constant, what is the value of k ?

Rationale

The correct answer is 4. It's given that $2x^2 + 5x - 12$ can be rewritten as $(2x-3)(x+k)$; it follows that $(2x-3)(x+k) = 2x^2 + 5x - 12$. Expanding the left-hand side of this equation yields $2x^2 + 2kx - 3x - 3k = 2x^2 + 5x - 12$. Subtracting $2x^2$ from both sides of this equation yields $2kx - 3x - 3k = 5x - 12$. Using properties of equality, $2kx - 3x = 5x$ and $-3k = -12$. Either equation can be solved for k. Dividing both sides of $-3k = -12$ by -3 yields $k = 4$. The equation $2kx - 3x = 5x$ can be rewritten as $x(2k-3) = 5x$. It follows that $2k - 3 = 5$. Solving this equation for k also yields $k = 4$. Therefore, the value of k is 4.